

SECTION 1

- 1 The solution(s) to the equation $|x - 2| = 2x + 1$ are?
- (A) $x = -3$
- (B) $x = -\frac{1}{3}$
- (C) $x = \frac{1}{3}$
- (D) $x = 3$
- 2 Which of the following is an expression for $\cos(A + B) - \cos(A - B)$?
- (A) $-2\cos A \cos B$
- (B) $-2\sin A \sin B$
- (C) $2\cos A \cos B$
- (D) $2\sin A \sin B$
- 3 Simplify $\frac{\sin(180^\circ + \theta)}{\cos(90^\circ - \theta)}$
- (A) -1
- (B) 1
- (C) $\tan \theta^\circ$
- (D) $\tan(90 - \theta)^\circ$
- 4 What is the exact value of $\sin(-105^\circ)$?
- (A) $-\frac{1 - \sqrt{3}}{2\sqrt{2}}$
- (B) $-\frac{1 + \sqrt{3}}{2\sqrt{2}}$
- (C) $\frac{\sqrt{3} + 1}{2\sqrt{2}}$
- (D) $\frac{\sqrt{3} - 1}{2\sqrt{2}}$
- 5 The solution to the inequality $\frac{6}{x(x - 2)} > 2$ is?
- (A) $-1 < x < 0$ or $0 < x < 2$
- (B) $x < -1$ or $2 < x < 3$
- (C) $x < -1$ or $0 < x < 2$
- (D) $-1 < x < 0$ or $2 < x < 3$

SECTION 2

QUESTION 1 (13 MARKS) Start a new writing booklet MARKS

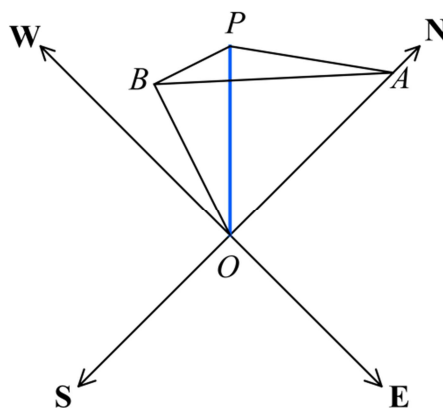
- a) Simplify $\frac{\sqrt{x^2}}{|x|}$ $x \neq 0$ 1
- b) (i) Factor $3^{n+1} + 3^n$, and 1
 (ii) hence simplify $\frac{3^{101} + 3^{100}}{4}$ 2
- c) Solve the inequality $\frac{|3-x|}{x^2+3} \geq 1$ 3
- d) Solve the inequality $7 - 3x < \frac{2}{x}$ for x . 3
- e) Sketch carefully $y = |x - 3| + |x + 2|$ 3

QUESTION 2 (13 MARKS) Start a new writing booklet MARKS

- a) Express $(3x - \frac{1}{x^3})^8$ in sigma notation. 2
- b) By considering $(2 + x - x^2)^4$ as $((2 + x) - x^2)^4$ 3
 Hence, or otherwise find the coefficient of x^4 .
- c) Show that the sum of the coefficients of the terms in $(2x + y)^5$ is 243. 2
- d) (i) If $\sqrt{3} \cos x + \sin x = R \cos(x - \alpha)$ where $R > 0$ and α is acute, show $R = 2$ and the angle $\alpha = 30^\circ$ 2
 (ii) Hence, or otherwise solve 2
 $\sqrt{3} \cos x + \sin x = 1$ for $0^\circ \leq x \leq 360^\circ$.
- e) Sketch the graph $y = 3 \cos \frac{x}{2}$ for $0^\circ \leq x \leq 360^\circ$. 2

QUESTION 3 (14 MARKS) Start a new writing booklet**MARKS**

- a) Express $\sqrt{\frac{4+4\cos 2x}{1-\cos 2x}}$ as a single trigonometric ratio. **3**
- b) Using the t -formulae, solve the equation $\sin 2x = \tan x$ for $0^\circ \leq x \leq 180^\circ$. **2**
- c) If $3\cos \theta + 2 = 0$ and $\tan \theta > 0$, find the exact value of:
- (i) $\tan(\theta - 180)^\circ$ **1**
- (ii) $\sin(\theta + 180)^\circ$ **1**
- d) Find the acute angle between the lines $y = -2x + 3$ and $x - 3y + 2 = 0$. **3**
Answer correct to the nearest degree.
- e) A vertical pole PO is 10 m high. Point A is due north of O and OA is the shadow of the pole when the angle of elevation to the sun is 67° . OB is the shadow cast by the pole when the direction of the sun is on a bearing of 340° and the sun's angle of elevation is 49° .



- (i) What is the size of $\angle AOB$? **1**
- (ii) Find the distance between A and B . Answer correct to 1 decimal place. **3**

END OF PAPER

Multiple Choice Answer

Section 1

C 2B 3 A 4 B 5 D.

5) $\frac{6}{x(x-2)} > 2 \quad x \neq 0$
 $x \neq 2$
 $x [x^2(x-2)]$

$$|x-2| = 2x+1$$

$$x-2 = 2x+1 \quad x-2 = -2x-1$$

$$-3 = x \quad 3x = 1$$

$$|-3-2| = -6+1 \quad x = \frac{1}{3}$$

(F)

$$|\frac{1}{3}-2| = 2 \times \frac{1}{3} + 1 \quad \checkmark \quad x = \frac{1}{3}$$

$$\frac{1}{3} = 1\frac{1}{3}$$

$$\cos(A+B) - \cos(A-B)$$

$$\cos A \cos B - \sin A \sin B$$

$$-(\cos A \cos B + \sin A \sin B)$$

$$-2 \sin A \sin B$$

(B)

$$\sin(180+\theta) = -\sin \theta$$

$$\cos(90-\theta) = \sin \theta$$

$$= -1$$

$$\sin(105)^\circ = -\sin(75)^\circ$$

$$= -\sin(45+30)$$

$$= -(\sin 45 \cos 30 + \sin 30 \cos 45)$$

$$= -\left(\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}}\right)$$

$$= -\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right) \quad B_{1/2}$$

$$6x(x-2) > 2x^2(x-2)^2$$

$$6x(x-2) - 2x^2(x-2)^2 > 0$$

$$2x(3(x-2) - x(x-2)^2) > 0$$

$$2x(x-2)(3 - x(x-2)) > 0$$

$$x(x-2)(3 - x^2 + 2x) > 0$$

$$x \neq 0, 2$$

$$-x(x-2)(x-3)(x+1) > 0$$

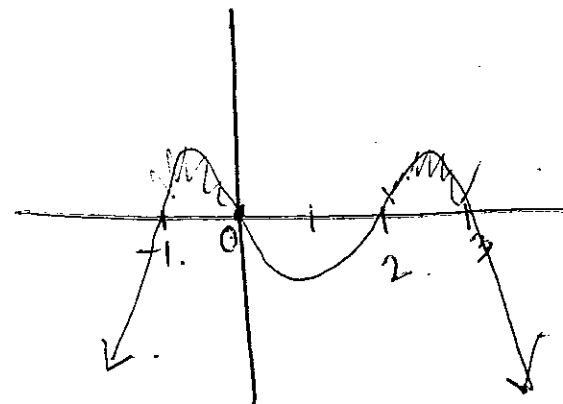
$$x = 0, 2, 3, -1$$

test

$$x = -1 \quad -1(-1)(-2) \cdot 2 > 0 \quad F$$

Solution

$$-1 < x < 0 \quad 2 < x < 3$$



Section 2 Q1.

a) $\frac{\sqrt{x^2}}{|x|} = 1$

b) $3^{n+1} + 3^n = 3^n(3+1)$

for correct substitution
 $\frac{3^{101} + 3^{100}}{4} = \frac{3^{100}(3+1)}{4} = 3^{100}$

c) $\frac{|3-x|}{x^2+3} \geq 1$

Case 1 Case 2

$|3-x| = 3-x \quad (x < 3)$ $|3-x| = -3+x \quad (x > 3)$

$\frac{3-x}{x^2+3} \geq 1$ $-3+x \geq x^2+3$

$3-x \geq x^2+3$ $0 \geq x^2-x+6$

$0 \geq x^2+x$ $0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

$0 \geq x(x+1)$

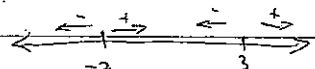
$0 \geq x(x+1)$

$0 \geq x(x+1)$

d) $y = |x-3| + |x+2|$

$|x-3| = x-3 \quad x > 3$
 $= -x+3 \quad x < 3$

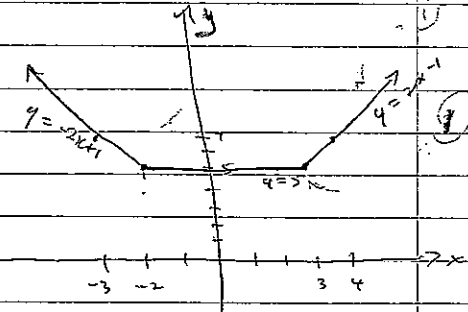
$|x+2| = x+2 \quad x > -2$
 $= -x-2 \quad x < -2$



$x > 3$ $x-3+x+2 = y$
 $2x-1 = y$

$x < 3$ $-x+3-x-2 = y$
 $-2x+1 = y$

$-2 < x < 3$ $-x+3+x+2 = y$
 $5 = y$



d) $7-3x < \frac{2}{x}$

$7x^2-3x^3 < 2x$

$0 < 3x^3-7x^2+2x$

$< x(3x^2-7x+2)$

$0 < x(3x-1)(x-2)$

$x=0 \quad x=\frac{1}{3} \quad x=2$

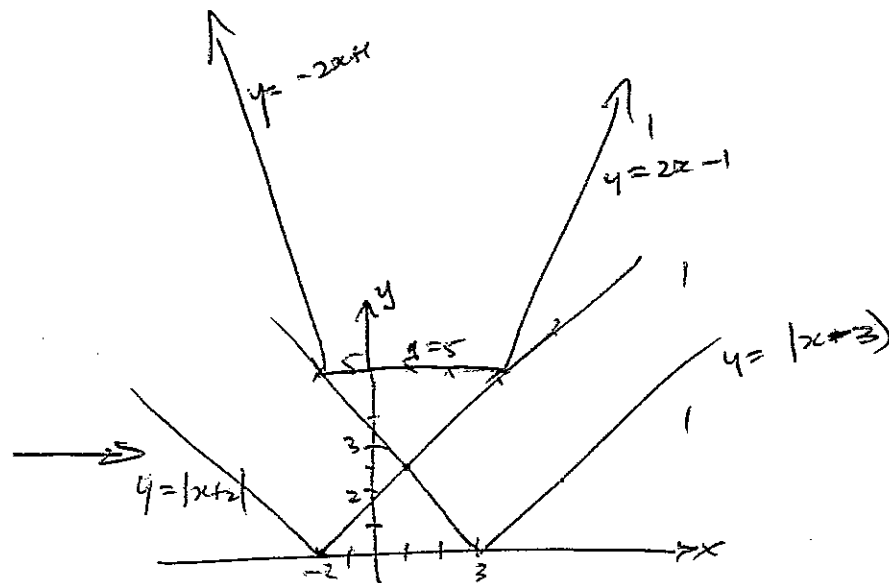
$x=1 \quad 7-3 < 2$

$0 < x < \frac{1}{3} \quad x > 2$

$0 < x < \frac{1}{3} \quad x > 2$

KP

$\sqrt{x^2} = |x|$



bold ans. ③

if graph only
 needs shape - (lines)
 - # for turning.

Ext 1 AP3 2017 Question 2

$$\begin{aligned}
 a) \left(3x - \frac{1}{x^3}\right)^8 &= \sum_{r=0}^8 \binom{8}{r} (3x)^{8-r} \left(-\frac{1}{x^3}\right)^r && \text{1mk.} \\
 &= \sum_{r=0}^8 \binom{8}{r} 3^{8-r} x^{8-r} (-1)^r x^{-3r} && \text{Substitutes appropriate terms into sigma notation, including boundaries of summation and binomial coefficient} \\
 &= \sum_{r=0}^8 \binom{8}{r} 3^{8-r} (-1)^r x^{8-4r} && \text{1mk.} \\
 &&& \text{Simplifies by combining } x \text{ terms}
 \end{aligned}$$

Alternate solution:

$$\begin{aligned}
 \left(3x - \frac{1}{x^3}\right)^8 &= \sum_{r=0}^8 \binom{8}{r} (3x)^r \left(-\frac{1}{x^3}\right)^{8-r} && \text{* Students may choose to write summation in reverse order (accepted with full marks if simplified accordingly).} \\
 &= \sum_{r=0}^8 \binom{8}{r} 3^r x^r (-1)^{8-r} (x^{-3})^{8-r} \\
 &= \sum_{r=0}^8 \binom{8}{r} 3^r (-1)^{8-r} x^r x^{-24+3r} \\
 &= \sum_{r=0}^8 \binom{8}{r} 3^r (-1)^{8-r} x^{4r-24} \\
 &= \sum_{r=0}^8 \binom{8}{r} 3^r (-1)^8 (-1)^{-r} x^{4r-24} \\
 &= \sum_{r=0}^8 \binom{8}{r} 3^r (-1)^r x^{4r-24} \\
 &= \sum_{r=0}^8 \binom{8}{r} (-3)^r x^{4r-24}
 \end{aligned}$$

$$\begin{aligned}
 b) (2+x-x^2)^4 &= [(2+x)-x^2]^4 \\
 &= \sum_{r=0}^4 \binom{4}{r} (2+x)^r (-x^2)^{4-r} \\
 &= \binom{4}{0} 1(x^8) + \binom{4}{1} (2+x)(-x^6) + \binom{4}{2} (4+4x+x^2)(x^4) \\
 &\quad + \binom{4}{3} (8+12x+6x^2+x^3)(-x^2) + \binom{4}{4} (16+32x+24x^2+8x^3+x^4)(1) \\
 &&& \text{1mk. correct expansion}
 \end{aligned}$$

$\therefore x^4$ term

$$\begin{aligned}
 &= \binom{4}{2} \times 4 \times (x^4) + \binom{4}{3} \times (6x) \times (-x^2) + \binom{4}{4} \times 1(x^4) \times (1) && \text{1mk. identification of relevant terms} \\
 &= 24x^4 + (-24x^4) + x^4
 \end{aligned}$$

$$= x^4 \quad \therefore \text{Coefficient is 1.} \quad \text{1mk. final answer}$$

$$\begin{aligned}
 c) (2x+y)^5 &= \binom{5}{0} (2x)^5 + \binom{5}{1} (2x)^4 y + \binom{5}{2} (2x)^3 y^2 + \binom{5}{3} (2x)^2 y^3 \\
 &\quad + \binom{5}{4} (2x) y^4 + \binom{5}{5} y^5 \\
 &= 1(32x^5) + 5(16x^4)y + 10(8x^3)y^2 + 10(4x^2)y^3 + 5(2x)y^4 + 1y^5 \\
 &&& \text{1mk. correct expansion}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Sum of coefficients} &= 32+80+80+40+10+1 && \text{1mk. evaluates coefficients} \\
 &= 243 \text{ as required.}
 \end{aligned}$$

$$(d)(i) \sqrt{3} \cos x + \sin x = R \cos(x - \alpha) \quad \{R > 0, 0^\circ < \alpha < 90^\circ\}$$

$$= R[\cos x \cos \alpha + \sin x \sin \alpha]$$

$$= R \cos \alpha (\cos x) + R \sin \alpha (\sin x)$$

$\therefore$ By comparison of coefficients:

$$R \cos \alpha = \sqrt{3} \quad \text{--- ①}$$

$$R \sin \alpha = 1 \quad \text{--- ②}$$

$$\text{①}^2 + \text{②}^2: R^2 = 4$$

$$R = 2 \quad \{R > 0\}$$

Sub R into ① & ②:

$$\text{①}: 2 \cos \alpha = \sqrt{3}$$

$$\text{②}: 2 \sin \alpha = 1$$

$$\cos \alpha = \frac{\sqrt{3}}{2}$$

$$\sin \alpha = \frac{1}{2}$$

$$\therefore \alpha = 30^\circ \quad \{0^\circ < \alpha < 90^\circ\}$$

Identifies coefficients from trigonometric expansion & equates with appropriate values from given expression

1 mk. combines simultaneous equations WITH WORKING (not sufficient for students to directly state $R = \sqrt{a^2 + b^2}$ or $\tan \alpha = \frac{b}{a}$ without justification)

$$(ii) \sqrt{3} \cos x + \sin x = 1 \quad \{0^\circ \leq x \leq 360^\circ\}$$

$$2 \cos(x - 30^\circ) = 1 \quad \text{from part (i)}$$

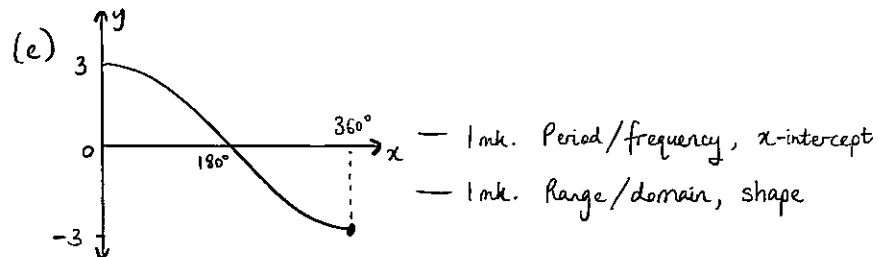
$$\cos(x - 30^\circ) = \frac{1}{2}$$

$$x - 30^\circ = 60^\circ \text{ or } 300^\circ \quad \{-30^\circ \leq x - 30^\circ \leq 330^\circ\}$$

$$x = 90^\circ \text{ or } 330^\circ$$

1 mk. solves for $x - 30^\circ$ or equivalent if (i) not used

1 mk. both solutions



Question 3

(a)

$$\frac{4 + 4 \cos 2x}{\sqrt{1 - \cos 2x}}$$

$$= \frac{4(1 + \cos 2x)}{\sqrt{1 - \cos 2x}} \quad \text{Factor}$$

$$= \frac{4(1 + 2\cos^2 x - 1)}{\sqrt{1 - (1 - 2\sin^2 x)}} \quad \checkmark$$

$$= \frac{2\cos^2 x}{2\sin^2 x}$$

$$= 2 \cot x \quad \checkmark$$

$$\text{OR } \frac{4(1 + \cos^2 x - \sin^2 x)}{\sqrt{1 - (\cos^2 x - \sin^2 x)}}$$

$$= 2 \frac{(1 - \sin^2 x) + \cos^2 x}{\sqrt{1 - \cos^2 x + \sin^2 x}} \quad \checkmark$$

$$= 2 \frac{\cos^2 x + \cos^2 x}{\sin^2 x + \sin^2 x}$$

$$= 2 \frac{2\cos^2 x}{2\sin^2 x}$$

$$= 2 \cot x \quad \checkmark$$

(b) $\sin 2x = \tan x$

$$\text{Let } t = \tan x$$

$$\therefore \sin 2x = \frac{2t}{1+t^2}$$

$$\frac{2t}{1+t^2} = t$$

$$2t = t + t^3$$

$$t^3 + t - 2t = 0$$

$$t^3 - t = 0$$

$$t(t^2 - 1) = 0 \quad \checkmark \quad \therefore x = 0^\circ, 45^\circ, 135^\circ, 180^\circ \quad \checkmark$$

$$t(t+1)(t-1) = 0$$

$$t = 0, t = \pm 1 \quad \checkmark$$

$$\tan x = 0$$

$$x = 0 \text{ or } 180^\circ$$

$$\tan x = 1$$

$$x = 45^\circ$$

$$\tan x = -1$$

$$x = 135^\circ$$

OR Long method.

$$\text{Let } t = \tan x/2$$

$$\tan x = \frac{2t}{1-t^2}$$

$$\sin 2x = 2 \sin x \cos x$$

$$= 2 \left(\frac{2t}{1+t^2} \right) \left(\frac{1-t^2}{1+t^2} \right)$$

$$\therefore \frac{2t}{1+t^2} \left(\frac{1-t^2}{1+t^2} \right) = \frac{2t}{1-t^2}$$

$$2t(1-t^2)^2 = t(1+t^2)^2$$

$$t[2(1-t^2)^2 - (1+t^2)^2] = 0$$

$$t[\sqrt{2}(1-t^2) + (1+t^2)][\sqrt{2}(1-t^2) - (1+t^2)] = 0$$

$$t[(\sqrt{2}+1) + t^2(1-\sqrt{2})][(\sqrt{2}-1) - t^2(\sqrt{2}+1)] = 0$$

$$t = 0, t^2 = \frac{\sqrt{2}+1}{\sqrt{2}-1}, t^2 = \frac{\sqrt{2}-1}{\sqrt{2}+1}$$

$$= (\sqrt{2}+1)^2 = (\sqrt{2}-1)^2$$

$$t = \pm(\sqrt{2}+1) \quad t = \pm(\sqrt{2}-1)$$

$$\tan \frac{x}{2} = 0 \quad \tan \frac{x}{2} = \sqrt{2}+1 \quad \tan \frac{x}{2} = (\sqrt{2}-1)$$

$$\frac{x}{2} = 0, \quad \frac{x}{2} = 67.5^\circ \text{ (using calc)} \quad \frac{x}{2} = 22.5^\circ$$

$$x = 135^\circ \quad \tan \frac{x}{2} = -(\sqrt{2}+1) \quad \text{Rejected out of domain}$$

$$x = 0$$

Check boundary

$$\text{Value } x = 180^\circ$$

$$0 < x \leq 180^\circ$$

$$0 \leq \frac{x}{2} \leq 90^\circ \quad \sin 2x = 0 \quad \tan 180^\circ = 0 \quad \therefore x = 180^\circ$$

$$x = 0, 45^\circ, 135^\circ, 180^\circ$$

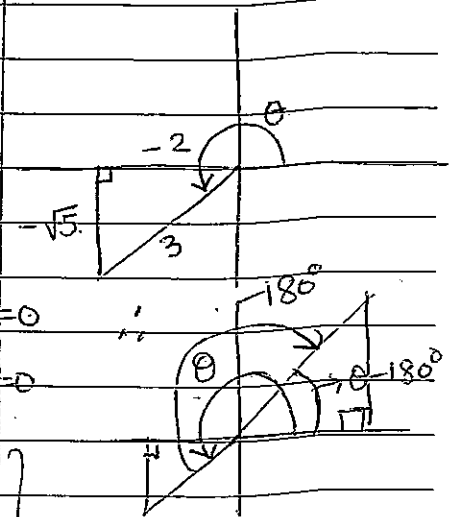
(c)

$$3 \cos \theta + 2 = 0$$

$$\cos \theta = -\frac{2}{3} < 0$$

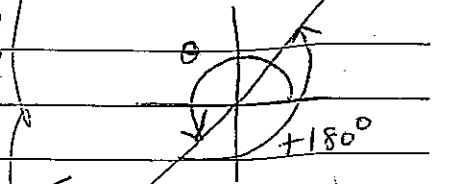
$$\tan \theta > 0$$

$\therefore \theta$ is in 3rd Quad.



$\theta - 180^\circ$ is in 1st Quad
 $\therefore \tan(\theta - 180^\circ) = \frac{\sqrt{5}}{2} \quad \checkmark$

ii) $\sin(\theta + 180^\circ)$
 $\theta + 180^\circ$ in 1st Quad



$$\sin(\theta + 180^\circ) = \frac{\sqrt{5}}{3} \quad \checkmark$$

$$d) y = -2x + 3 \quad x - 3y + 2 = 0$$

$$m_1 = -2 \quad m_2 = \frac{1}{3} \quad \left. \vphantom{\begin{matrix} m_1 \\ m_2 \end{matrix}} \right\} \checkmark$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-2 - \frac{1}{3}}{1 - \frac{2}{3}} \right|$$

$$= 7$$

$$\therefore \theta = 82^\circ \checkmark$$

including absolute value sign
should clearly differentiate from prev. step with $\tan \theta =$ and $\theta =$

$$OA = \frac{10}{\tan 67^\circ} \text{ or } 10 \cot 67^\circ$$

$$\text{or } 10 \tan 27^\circ$$

OA & OB expressions $\checkmark$

$$AB^2 = OA^2 + OB^2 - 2OA \times OB \cos 20^\circ$$

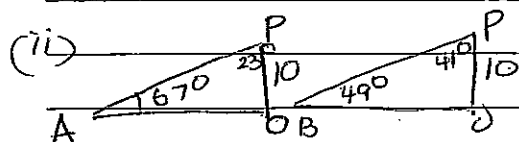
$$= (10 \tan 27^\circ)^2 + (10 \tan 41^\circ)^2$$

$$- 2(10 \tan 27^\circ)(10 \tan 41^\circ) \times \cos 20^\circ$$

$$\approx 4.9 \text{ m.} \checkmark$$

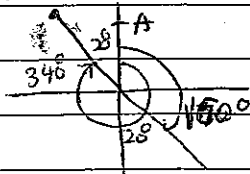
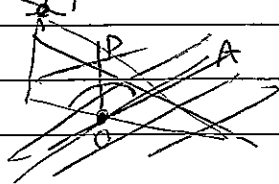
3(e) As per the diagram,

$$(i) \angle AOB = 360^\circ - 340^\circ = 20^\circ$$



$$OB = \frac{10}{\tan 49^\circ} \text{ or } 10 \cot 49^\circ \text{ or } 10 \tan 41^\circ$$

(e) As per wordings



$$(i) \angle AOB = 160^\circ \checkmark$$

$$OA =$$

$$OB =$$

$\checkmark$ See previous method

$$AB^2 = (10 \tan 41^\circ)^2 + (10 \tan 23^\circ)^2 - 2(10 \tan 41^\circ)(10 \tan 23^\circ) \cos 160^\circ$$

$$= 162.9313279 \dots$$

$$AB \approx 12.8 \text{ m.} \checkmark$$